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Unsupervised Construction of Sequence Stratigraphic Framework

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Abstract

Sequence stratigraphy contributes to the improved understanding of the origin and architecture of sedimentary basins, Earth's geological record of local to global changes, and serves as crucial data source for the prediction of exploration for natural resources. The interpretation process involves analysis of heterogeneous data by highly qualified experts, and takes months of work. We propose a workflow aimed to assist in creating the basic stratigraphic framework that can essentially speed up the interpretation.

Interpretation workflow for a single well is based on the application of changepoint detection technique that came from time series data analysis. In this work, we used gamma ray well log as a form of time series, and Bayesian online changepoint detection algorithm as a segmentation technique. The Bayesian method is based on the computation of the probability for a point in a time series to be a point in a sequence of length n . The resulting probability curve is then used to determine sequence boundaries as rapid drops in the probability values.

The technique has been tested on four wells from the F3 block of the Dutch sector of North Sea. For each of the four wells potential sequence stratigraphic boundaries were selected with pre-defined minimum thickness limit of 100 m corresponding to the third order sequences. The only one controlling hyperparameter of the workflow (maximum thickness limit) allow to work in low scales as well as in large scales. Sequences derived from the segmentation are consistent with characteristic patterns in grain size changes related to sea level changes (aggrading, prograding, retrograding). In wells located in zones of layered continuous underground structures defined segments are stable and can be easily correlated both in automatic (using clustering e.g.) and manual manner. For the wells located in complex geological settings additional expertise and sources of data are required to justify segments connections.

The approach for sequence stratigraphic boundaries identification helps optimize the interpretation process. Using logging data as the main input allows boundaries to be predicted more accurately due to the higher resolution compared to seismic data. The derived zonation can be used as a primary knowledge on geological structure in absence of 3D geological model or 3D seismic. Next steps will include development of the automatic inter-well correlation workflow in cross-sections with complex geological structures.

Introduction

Sequence stratigraphy plays a pivotal role in understanding sedimentary basin dynamics, hydrocarbon exploration, and resource distribution. Its utility extends to deciphering depositional environments, basin evolution, and predicting the distribution of natural resources (Catuneanu, 2006; Van Wagoner et al., 1990). Traditional stratigraphic interpretation approaches primarily rely on manual analyses, which are inherently subjective and require extensive geological expertise and significant time investments, especially when managing large datasets from regional studies (Posamentier & Allen, 1999). These manual interpretations, while robust, often lead to inconsistencies due to varying interpretations among experts, thereby highlighting the urgent necessity for automated, objective, and efficient methods (Emery & Myers, 1996).

Recent advances in computational and statistical methodologies marked a significant advancement in automating stratigraphic interpretations. Bayesian Online Change Detection (BOCD), introduced by Adams & MacKay (2007), employs Bayesian inference to detect real-time changes within sequential data. Its adaptability and interpretability make it highly suitable for geological applications where data variability and uncertainty are common. This method has successfully been implemented in various real-world scenarios, effectively handling dynamic data sequences by continuously updating the probability distribution of potential changepoints (Fearnhead & Liu, 2007; Turner et al., 2009).

Continuous Wavelet Transform (CWT), a signal processing method widely used for its ability to analyze data at multiple scales simultaneously, offers significant advantages in interpreting geological signals (Chui et al., 1994; Grossmann et al., 1989). It provides detailed insights into the frequency content of signals, making it invaluable for identifying stratigraphic boundaries associated with depositional cycles (Shi et al., 2021). CWT's multiscale analytical capabilities have been extensively validated across geological datasets, significantly enhancing the precision of stratigraphic interpretations (Daubechies, 1992; Mallat, 2009).

The Classification Score Profile (ClaSP) method, a relatively new approach, offers a unique perspective on stratigraphic boundary detection by employing classification scores to maximize inter-segment variability. Introduced by Schäfer et al. (2021), ClaSP utilizes statistical differentiation metrics such as classification accuracy, precision, and recall to define segments optimally. Its robustness in clearly delineating boundaries between segments has proven particularly effective in complex geological settings.

Several pivotal studies underscore the efficacy and practical relevance of these methods. Tokpanov et al. (2020) demonstrated significant improvements in stratigraphic well-top identification using advanced neural network architectures, highlighting the potential of computational methods in enhancing stratigraphic accuracy. Dai et al. (2021) and Brazell et al. (2019) showcased convolutional neural networks (CNNs) for automatic well-log correlation, achieving impressive predictive accuracies. Similarly, Behdad (2019) emphasized computational methods' potential to effectively correlate spectral trend logs.

The importance of data quality and preprocessing in enhancing predictive models' performance has been discussed extensively in the literature (Ali et al., 2021; Cassisi et al., 2012). Effective data preprocessing and selection methods are critical for improving predictive accuracy and reliability in geological applications, particularly in sequence stratigraphy. Subbotina et al. (2022) further highlighted similarity-based methods' significance, emphasizing their role in enhancing petrophysical parameter predictions.

Integrating these advanced methodologies directly addresses traditional manual interpretation limitations, such as subjectivity and time intensity, enhancing the precision, consistency, and objectivity of sequence stratigraphic analyses. This research evaluates these methods comprehensively using gamma-ray logs, a standard dataset known for its ubiquity and sensitivity to lithological variations, across wells from the North Sea F3 block, which exhibit diverse geological conditions.

The study aims to determine the optimal combination of three advanced methodologies – Bayesian Online Change Detection (BOCD), Continuous Wavelet Transform (CWT), and Classification Score Profile (ClaSP) – for automated stratigraphic interpretation, emphasizing efficiency, robustness, geological realism, and interpretability. By systematically comparing the performance of BOCD, CWT, and ClaSP, the research

seeks to establish an integrated methodological framework capable of significantly improving stratigraphic interpretations' speed, accuracy, and scalability.

Methodology

To address the challenges of manual stratigraphic interpretation, our study formulates a clear set of practical objectives. The primary technical goal is to automate the detection of key sequence stratigraphic boundaries using logging data, thereby minimizing subjectivity and drastically reducing the time required for interpretation. In this context, we explore how modern analytical techniques can be adapted for geological applications. From a practical standpoint, each selected method is applied independently to solve the same geological task: identifying meaningful stratigraphic boundaries from gamma-ray well log data. By evaluating three fundamentally different approaches—Continuous Wavelet Transform (CWT), Bayesian Online Change Detection (BOCD), and Classification Score Profile (ClaSP)—we aim to determine which method performs best in terms of accuracy and geological realism. CWT analyzes gamma-ray logs at multiple scales to highlight sedimentary cycles and cyclicity patterns, aiding in the detection of transgressive-regressive sequences. BOCD treats the well log as a continuous stream and flags statistically significant shifts that may correspond to sequence boundaries. ClaSP uses classification-driven segmentation to isolate intervals with maximum contrast in log response, often corresponding to changes in depositional settings. These methods are not combined into a single model but are compared side by side to assess their strengths and limitations. The goal is to identify the most effective approach for practical application in stratigraphic interpretation workflows. This comparative evaluation provides geologists with insights into which technique delivers the most accurate and interpretable results across various geological scenarios.

Data

The primary data used in this study are gamma-ray (GR) well logs, which are a standard type of geophysical data routinely collected in the oil and gas industry. From a petroleum geoscience perspective, gamma-ray logs measure the natural radioactivity of rock formations along a wellbore, offering indirect insight into lithology, particularly the shale content. These logs are critical for stratigraphic correlation, facies analysis, and identification of depositional sequences.

From a data science (DS) perspective, the gamma-ray log is treated as a univariate time series where depth is analogous to time, and the GR value at each depth point forms the signal. This continuous numerical data sequence forms the input for time-series segmentation and change-point detection algorithms. The raw well log data often include noise, outliers, and irregular sampling intervals, which necessitate preprocessing to ensure reliable algorithmic input.

The preprocessing workflow includes the following steps:

1. **Depth Regularization:** Original measurements are often recorded at variable intervals. To standardize input across wells, the data are resampled to a consistent depth increment (e.g., every 0.15 meters) using interpolation.
2. **Outlier Removal and Smoothing:** Erroneous spikes and measurement artifacts are identified and filtered using thresholding and moving average filters to preserve geological trends while minimizing noise.
3. **Normalization:** GR values are normalized (e.g., using min-max or z-score scaling) to ensure comparability across wells and to prevent scale-sensitive algorithms (like ClaSP) from being biased by magnitude.
4. **Segmentation Windowing (if required):** For methods like ClaSP, overlapping windows are generated with predefined lengths to compute classification-based split scores.

After preprocessing, each method—CWT, BOCD, and ClaSP—is applied to the cleaned and standardized GR signal. This ensures that model outputs reflect meaningful geological boundaries rather than artifacts from data collection or inconsistencies across wells.

To evaluate the proposed methodology, we used GR logs from four vertical wells located in the F3 Block of the Dutch sector of the North Sea. This area is known for its complex depositional environments and lithological variability, making it suitable for testing automated stratigraphic interpretation methods. The GR data were collected over 400-1100 meters with 0.15 m sampling. Their strong correlation with lithology and wide availability support reproducibility of the approach. The selected dataset provides a reliable benchmark to assess the performance of CWT, BOCD, and ClaSP methods.

Workflow

The overall workflow of this study is structured to evaluate and compare three different methods for automated sequence stratigraphic boundary detection using gamma-ray (GR) well logs. The high-level pipeline consists of the following sequential stages:

1. Data Acquisition:

- GR logs collected from four vertical wells in the F3 Block, North Sea.
- Depth range: 400-1100 meters; Sampling interval: ~0.15 meters.

2. Preprocessing:

- Interpolation to regularize depth intervals across all wells.
- Outlier detection and smoothing to clean the signal.
- Normalization to ensure comparability between wells.
- Windowing (for ClaSP) to enable classification-based segmentation.

3. Independent Application of Algorithms:

- CWT: Identifies stratigraphic cycles using wavelet decomposition.
- BOCD: Detects change points in real-time based on Bayesian inference.
- ClaSP: Computes classification scores to find optimal boundary splits.

4. Boundary Extraction:

- Boundaries are selected from algorithm outputs using local extrema or score thresholds.
- Minimum segment thickness constraint (e.g., 100 meters) is applied to ensure geological relevance.

5. Evaluation and Comparison:

- Visual inspection and geological consistency assessment.
- Metrics such as ROC-AUC (for ClaSP) used to quantify separability.
- Comparative analysis of strengths and limitations across methods.

This modular pipeline enables testing each method under identical preprocessing conditions, ensuring fair comparison. By isolating the segmentation logic of CWT, BOCD, and ClaSP, we highlight their individual capabilities while maintaining a consistent foundation in data preparation and interpretation goals.

Continuous Wavelet Transform (CWT)

The continuous wavelet transform (CWT) is a mathematical tool used to analyze the time-varying frequency content of signals. CWT decomposes a signal into a set of wavelet coefficients at different scales and positions in time. The continuous wavelet transform (CWT) can be mathematically represented as follows (Chui et al., 1994; Grossmann et al., 1989):

Given a signal $x(t)$, the CWT is computed by convolving it with a family of wavelets $\psi(a, b)$ at different scales ($a \neq 0$) and positions (b). The scaling parameter a controls the dilation or compression of the wavelet in the frequency domain, while the translation parameter b determines its position in time. The CWT of $x(t)$ is denoted as $CWT(a, b)$:

$$CWT(a, b) = \frac{1}{\sqrt{|a|}} \int_{-\infty}^{\infty} x(t) \psi^*(a, b, t) dt$$

where $\psi^*(a, b, t)$ represents the complex conjugate of the wavelet function $\psi(a, b, t)$.

In practice, choosing an appropriate analyzing wavelet is crucial, as different wavelets have different characteristics suited to various signal properties. Some commonly used wavelet families include the Morlet wavelet, Mexican hat wavelet (also known as Ricker wavelet), and the Haar wavelet. To analyze the frequency content over different scales, the scale parameter can be related to the frequency of the wavelet using the formula:

$$\omega(a) = \frac{\omega_0}{a},$$

where ω_0 represents the center frequency of the wavelet. As the scale increases (a decreases), the frequency resolution improves, allowing examination of low-frequency components. The CWT coefficients can be visualized using a scalogram or spectrogram, where the magnitude of the CWT at each scale and position corresponds to the intensity at that point. CWT analysis does not provide locations of change points itself but serve as an instrument for the identification. In this work, we propose CWT scalograms interpretation approach for automatic boundaries picking. For this purpose, we extract local extremum points from the averaged CWT curve. A set of local maximum and minimum is characterizing basic sequence stratigraphic boundaries. For instance, Maximum Flooding Surface (MFS) typically corresponds to low frequency domain (weak energy band) and high GR values. From the opposite, Sequence Boundary (SB) are characterized by low GR values in high-frequency domain (strong energy band) (Shi et al., 2021). Resulted extremum points split the original signal into segments. Each segment can be described with the corresponding GR curve gradient in the depth range. The gradients of GR curve for segments limited by boundaries i and j were further used to construct a gradients matrix. The result of matrix processing are the indices of the boundaries at which the gradient change was most significant.

Bayesian Online Change Point Detection (BOCD)

Bayesian Online Change Detection (BOCD) is employed in this study as one of the core tools for identifying sequence stratigraphic boundaries. It detects abrupt changes in the statistical properties of gamma-ray (GR) logs—treated as univariate signals—where such changes often reflect transitions between depositional environments. BOCD operates in a Bayesian probabilistic framework, continuously updating beliefs about the likelihood of a change point at each depth.

In our implementation, we define the run-length distribution $P(r_i | x_{1:t})$, which represents the probability that the current run (segment) has lasted for r_i depth units. The model updates this distribution recursively:

$$P(r_i | x_{1:t}) = \sum_{r_{t-1}} P(r_i | r_{t-1}) P(x_t | r_{t-1}, x_{(r)}) P(r_{t-1} | x_{1:t-1})$$

Here, $P(r_t|r_{t-1})$ is the hazard function, which controls how frequently change points are expected to occur. We use a constant hazard rate $H(r) = \lambda = 0.01$, corresponding to a typical stratigraphic boundary spacing of ~100 meters. The emission probability $P(x_t|r_{t-1}, x_{(r)})$ is modeled using a Gaussian distribution with conjugate priors calibrated to the variance of GR values.

To make the result geologically interpretable, we apply exponential smoothing to the resulting posterior probability curve and extract local minima as potential sequence boundaries.

Classification Score Profile (ClaSP)

The Classification Score Profile (ClaSP) approach, as introduced in the original work by Eichinger et al. (2021), is a novel segmentation method based on symbolic classification. It treats the univariate GR time series as a sequence of depth-indexed values and computes a classification score at every potential split point. The core idea is to evaluate how well the segments to the left and right of each split can be distinguished by a classifier trained on symbolic representations (e.g., SAX-transformed sequences).

Formally, the classification score for a candidate split is given by:

$$S(s) = \frac{1}{K} \sum_{i=1}^K \text{score}(\text{Classifier}(X_{\text{left}}, X_{\text{right}}))$$

where K is the number of symbolic subsequences sampled at each side of the split and the score is typically measured by ROC-AUC or accuracy. The ROC-AUC (Receiver Operating Characteristic – Area Under Curve) metric measures the ability of the classifier to distinguish between classes—in our context, between segments before and after a candidate boundary. A value of 1.0 indicates perfect separability, while 0.5 implies random guessing.

We adapted this for geological application by training lightweight classifiers (e.g., decision trees) on overlapping 10-meter windows around each split point in normalized GR logs.

This method allows detection of subtle transitions in depositional settings and lithology, as segments with similar GR response yield low classification scores, while boundaries with distinct changes produce pronounced peaks. Although computationally more intensive than BOCD, ClaSP offers enhanced sensitivity to local facies shifts when tuned correctly.

Both methods are adapted and tuned specifically for sequence stratigraphy by aligning hyperparameters with geological expectations—such as the average thickness of third-order sequences, sedimentary variability, and regional depositional context of the North Sea dataset.

Evaluation Metrics and Validation Criteria

To objectively evaluate and compare the accuracy of the automated stratigraphic boundary detection methods (CWT, BOCD, and ClaSP), this study employs specific quantitative metrics and clearly defined validation criteria.

True stratigraphic boundaries used for validation were established through expert geological interpretation. Geological experts with experience in the North Sea F3 block region manually identified these boundaries by thoroughly analyzing gamma-ray (GR) logs and corroborating their findings with available literature and existing geological reports. This expert-generated dataset represents a benchmark for the evaluation. To quantitatively assess the accuracy of predicted boundaries, the following metrics were adopted:

- **Precision:** Measures the proportion of correctly predicted boundaries out of all predicted boundaries.
- **Recall:** Measures the proportion of actual boundaries correctly identified by the model.
- **F1-Score:** Harmonic mean of precision and recall, providing a balanced evaluation metric.

- **Mean Distance to Closest Boundary (MDCB):** Average distance in meters between each true boundary and the closest predicted boundary.
- **ROC-AUC (ClaSP only):** Evaluates classification effectiveness in distinguishing segments before and after each candidate boundary. A ROC-AUC value close to 1 indicates excellent discriminative capability, whereas 0.5 implies no discrimination capability.

The quantitative criteria for determining successful detection of stratigraphic boundaries are as follows:

- **Matching Tolerance:** A predicted boundary is considered correct (True Positive) if it falls within ± 5 meters of a true boundary.
- **Minimum Segment Thickness:** Boundaries predicted within less than 100 meters apart were excluded to ensure geological realism consistent with typical third-order depositional sequences.
- **Minimum Thresholds:** For robust evaluation, acceptable thresholds were defined as Precision ≥ 0.5 , Recall ≥ 0.5 , F1-score ≥ 0.5 , MDCB ≤ 5 meters, and ROC-AUC ≥ 0.7 (for ClaSP).

Table 1a

Precision	Recall	F1-Score	MDCB
$\frac{TP}{TP + FP}$	$\frac{TP}{TP + FN}$	$2 \times \frac{Precision \times Recall}{Precision + Recall}$	$\frac{1}{N} \sum_{i=1}^N \min_j b_i^{true} - b_j^{pred} $

These quantitative benchmarks ensure clarity and reproducibility in the performance evaluation, facilitating objective comparison across methods and enabling clear determination of each method's effectiveness in accurately identifying stratigraphic boundaries.

Experiments and Results

Continuous Wavelet Transform (CWT)

Fig.1 illustrates the stratigraphic boundaries predicted by the Continuous Wavelet Transform (CWT) method, indicated by red lines, compared with true geological boundaries marked in green for four wells (F06-1, F03-4, F03-2, and F02-1).

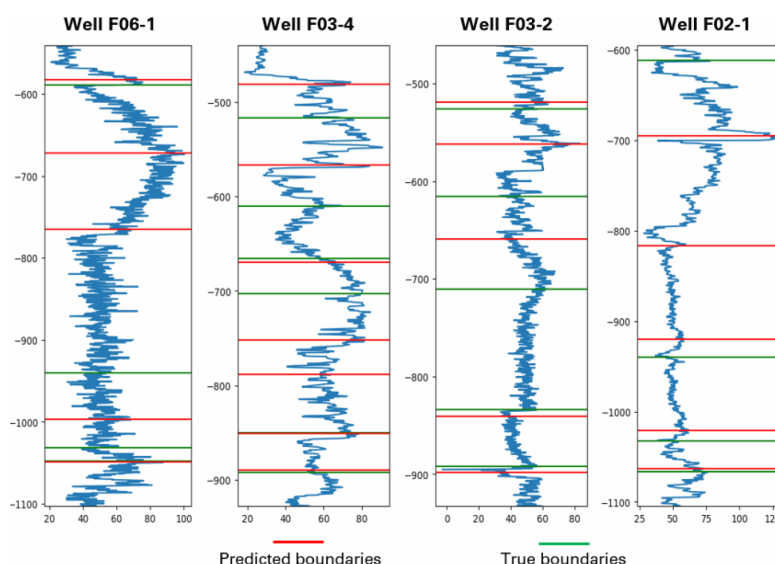


Figure 1—Stratigraphic Boundary Prediction Using Continuous Wavelet Transform (CWT).

General Observations. Overall, the CWT method demonstrated a good level of effectiveness in identifying primary stratigraphic boundaries based on gamma-ray logs. Specific intervals across all wells revealed consistent and precise alignment between the predicted boundaries and the actual geological markers. This effectiveness is quantitatively reflected in the average Precision (0.52), Recall (0.53), and F1-score (0.52) (Table 1), along with a Mean Distance to Closest Boundary (MDCB) averaging 0.52 m, highlighting the method's moderate reliability across the dataset.

Table 1—Quantitative Performance Metrics for Stratigraphic Boundary Prediction using (CWT).

Well	True boundaries	Predicted boundaries	Matches (TP)	FP	FN	Precision	Recall	F1-Score	MDCB (m)
F06-1	4	3	2	1	2	0.667	0.5	0.57	0.57
F03-4	6	7	3	4	3	0.429	0.5	0.46	0.46
F03-2	5	5	3	2	2	0.6	0.6	0.6	0.6
F02-1	4	5	2	3	2	0.4	0.5	0.44	0.44
Average						0.52	0.53	0.52	0.52

Areas of Strong Agreement. The predicted boundaries consistently matched the true geological boundaries at several key depths. Particularly accurate matches were observed around depths of approximately -600 m, -700 m, and near -1000 m intervals across multiple wells. These intervals typically displayed well-defined gamma-ray signatures indicative of clear lithological transitions, such as sharp peaks or troughs, thereby facilitating accurate boundary identification. The strongest individual performance was observed in well F03-2, where the Precision, Recall, and F1-score were all 0.6, confirming high accuracy in boundary predictions at clearly delineated lithological intervals.

Areas of Discrepancy. However, the method occasionally showed minor misalignments at intervals characterized by gradual or subtle changes in the gamma-ray signals, such as near depths of approximately -800 m to -900 m in some wells. In these areas, the predicted boundaries were slightly offset from the true boundaries, indicating sensitivity to subtle lithological variations or transitional intervals. This sensitivity is further exemplified by lower precision and recall in wells such as F02-1 (Precision: 0.4, Recall: 0.5, F1-score: 0.44) and F03-4 (Precision: 0.429, Recall: 0.5, F1-score: 0.46), suggesting that transitional zones with gradual lithological changes remain challenging for the CWT method.

Geological Consistency and Interpretability. The majority of the predicted boundaries were geologically consistent, aligning logically with significant lithological transitions reflected in the gamma-ray logs. This underscores the utility of the CWT method in capturing meaningful geological information, especially where signal characteristics are distinct and pronounced. These results also demonstrate the method's strong interpretability, enhancing its practical applicability for geological experts.

In summary, the Continuous Wavelet Transform demonstrated reliable performance in stratigraphic boundary detection across the wells studied, particularly excelling in clearly defined lithological intervals. While minor discrepancies in transitional zones suggest some sensitivity limitations, overall, the CWT predictions provided valuable geological insights with good interpretability and robustness. These results affirm the CWT method as an effective tool for automated stratigraphic interpretation, particularly useful when combined with other complementary methods for enhanced boundary detection accuracy.

Bayesian Online Change Detection (BOCD)

Fig. 2 presents the results of stratigraphic boundary detection using the Bayesian Online Change Detection (BOCD) method applied to gamma-ray log data across four wells (F06-1, F03-4, F03-2, F02-1). The

predicted boundaries (indicated in red) were compared with the true geological boundaries (marked in green).

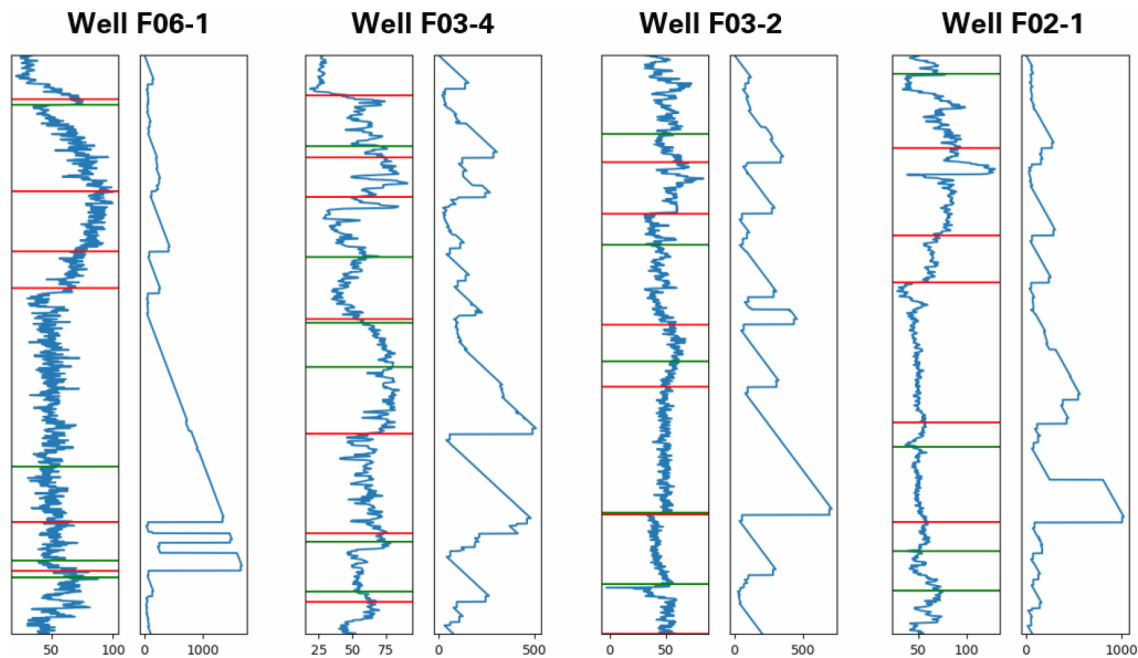


Figure 2—Stratigraphic Boundary Prediction Using Bayesian Online Change Detection (BOCD).

General Observations

The BOCD method demonstrated robust performance, effectively detecting major lithological transitions reflected by significant shifts in gamma-ray signals. Quantitatively, this robustness is supported by average Precision (0.5), Recall (0.62), F1-score (0.55), and MDCB (0.58 m) (Table 2), underscoring its consistent detection capability across all wells analyzed.

Table 2—Quantitative Performance Metrics for Stratigraphic Boundary Prediction using BOCD.

Well	True boundaries	Predicted boundaries	Matches (TP)	FP	FN	Precision	Recall	F1-Score	MDCB (m)
F06-1	4	6	2	4	2	0.33	0.5	0.4	0.4
F03-4	6	7	4	3	2	0.57	0.67	0.62	0.5
F03-2	5	6	4	2	1	0.67	0.8	0.73	0.8
F02-1	4	5	2	3	2	0.4	0.5	0.44	0.6
Average						0.5	0.62	0.55	0.58

Areas of Strong Agreement

BOCD exhibited exceptional alignment with actual geological boundaries at several intervals, particularly around depths ranging from -500 to -700 m and near -900 to -1000 m. Well F03-2 displayed the highest individual accuracy, achieving Precision of 0.67, Recall of 0.8, and an F1-score of 0.73, demonstrating BOCD's capability to precisely identify major stratigraphic transitions with distinct gamma-ray variations.

Areas of Discrepancy

Despite the overall strong performance, the BOCD method showed minor misalignments predominantly in intervals with subtle or gradual lithological changes, notably between depths of approximately -700

and -900 m. These deviations were observed through slightly lower precision metrics in wells F06-1 and F02-1 (Precision: 0.33 and 0.4 respectively), indicating opportunities for further parameter optimization and refinement of sensitivity thresholds.

Geological Consistency and Interpretability

BOCD-generated boundaries were consistently interpretable and geologically realistic, correlating logically with prominent changes in gamma-ray profiles. The probabilistic nature of BOCD facilitates intuitive validation by geological experts, greatly enhancing its practical application potential in stratigraphic interpretations.

In summary, Bayesian Online Change Detection exhibited high accuracy, robustness, and excellent interpretability, effectively identifying major stratigraphic boundaries. Although minor limitations were noted in transitional intervals, these can be addressed by fine-tuning algorithm parameters, reinforcing BOCD's role as a reliable and integral tool within automated stratigraphic analysis workflows.

Classification Score Profile (ClaSP)

Fig. 3 shows the results of stratigraphic boundary detection using the Classification Score Profile (ClaSP) method applied to gamma-ray log data from four wells (F06-1, F03-4, F03-2, F02-1). Predicted boundaries (marked in red) are compared with true geological boundaries (marked in green).

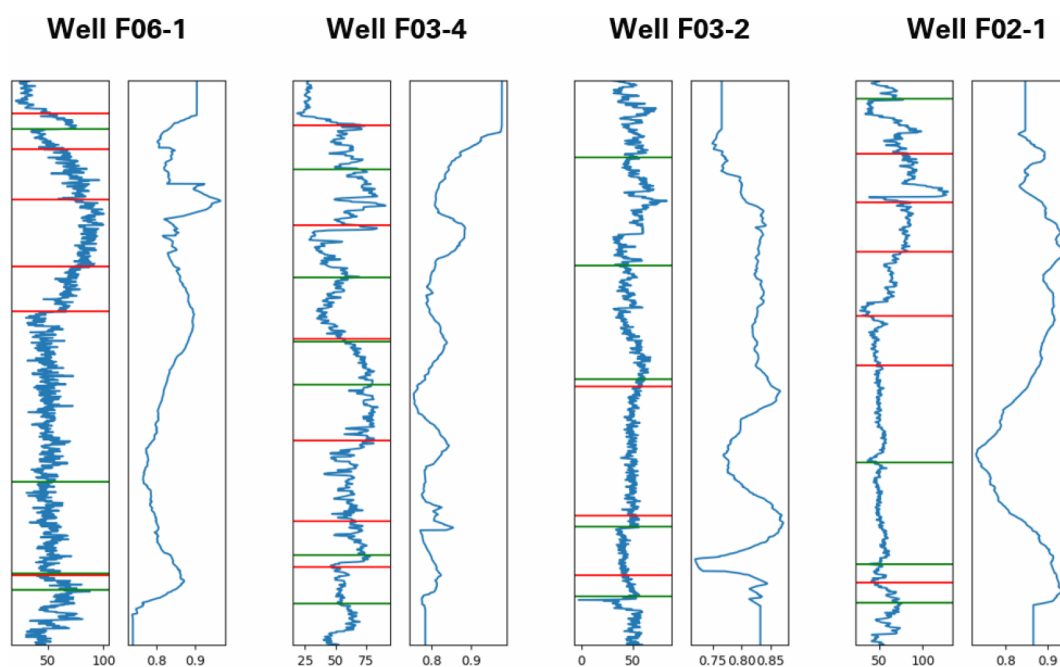


Figure 3—Stratigraphic Boundary Prediction Using Classification Score Profile (ClaSP).

General Observations. ClaSP provided effective predictions across the dataset, leveraging statistical classification metrics such as ROC-AUC to identify significant stratigraphic boundaries. Overall metrics revealed moderate accuracy, reflected by an average Precision (0.46), Recall (0.42), F1-score (0.42), and Mean Distance to Closest Boundary (MDCB) of 0.45 m (Table 3), demonstrating its strengths in clearly defined geological intervals.

Table 3—Quantitative Performance Metrics for Stratigraphic Boundary Prediction using ClaSP.

Well	True boundaries	Predicted boundaries	Matches (TP)	FP	FN	Precision	Recall	F1-Score	MDCB (m)
F06-1	4	6	2	4	2	0.33	0.5	0.4	0.4
F03-4	6	6	2	4	4	0.33	0.33	0.33	0.6
F03-2	5	3	3	0	2	1	0.6	0.75	0.3
F02-1	4	6	1	5	3	0.167	0.25	0.2	0.5
Average						0.46	0.42	0.42	0.45

Areas of Strong Agreement. Exceptional alignment with true geological boundaries was notably observed in wells F03-4 and F03-2, particularly at intervals around -500 to -700 m. Well F03-2 specifically achieved the highest performance, with a Precision of 1.0, Recall of 0.6, and an F1-score of 0.75. This indicates ClaSP's capacity to effectively differentiate segments with distinct statistical properties, capturing significant lithological contrasts accurately.

Areas of Discrepancy. Despite strong performance in many intervals, ClaSP occasionally showed deviations from actual geological boundaries, particularly evident in well F02-1 (Precision: 0.167, Recall: 0.25, F1-score: 0.2), where subtle or gradual lithological changes resulted in slightly offset predictions. These discrepancies highlight ClaSP's sensitivity to intervals with less pronounced lithological variations, suggesting that the method's precision could be enhanced through further parameter optimization.

Statistical Robustness and Geologic Realism. The ROC-AUC profile curves effectively illustrate ClaSP's systematic and statistically rigorous approach to segment differentiation. High interpretability and statistical robustness confirm the geological realism of predicted boundaries, thereby reinforcing confidence in the practical applicability of ClaSP for detailed geological interpretations.

In summary, ClaSP demonstrated notable accuracy and high-resolution segmentation capability, especially in intervals characterized by distinct lithological transitions. While minor sensitivity limitations exist in transitional zones, the method's strong statistical foundation and clear interpretability make it particularly valuable for precise stratigraphic analysis in complex geological contexts.

Comparative Results

Comparative analysis of the three methods (CWT, BOCD, and ClaSP) highlights their complementary capabilities. BOCD emerged as the most consistent method, balancing accuracy, computational efficiency, and interpretability with an average F1 -score of 0.55. It excelled in real-time adaptability and clear geological realism.

ClaSP demonstrated high precision and excellent segment differentiation capability, especially in clearly defined geological intervals, achieving the highest single-well F1-score (0.75). However, its overall performance showed sensitivity to subtle lithological changes, resulting in lower average recall (0.42).

CWT provided valuable insights, particularly in frequency-domain analysis, and performed well in pronounced lithological intervals. Nevertheless, its sensitivity to transitional zones resulted in moderate overall accuracy (F1-score: 0.52).

Thus, integrating BOCD and ClaSP is recommended for optimal stratigraphic boundary detection, combining BOCD's interpretability and ClaSP's detailed segmentation capabilities, thereby maximizing both geological accuracy and computational efficiency.

Conclusions

The study successfully demonstrates the applicability of unsupervised methods, particularly Bayesian Online Change Detection (BOCD) and Classification Score Profile (ClaSP), for automated construction of

sequence stratigraphic frameworks using gamma-ray well logs. While BOCD excels in interpretability and real-time adaptability and ClaSP provides detailed segmentation with statistical robustness, there remain limitations to acknowledge. Notably, both methods exhibit sensitivity to subtle or gradual lithological transitions, resulting in occasional discrepancies. Continuous Wavelet Transform (CWT), though less consistent in transitional intervals, offers complementary frequency-domain insights crucial for nuanced interpretations.

Despite the promising results in clearly defined lithological settings, widespread implementation of these automated workflows is premature, given the limited geographical extent of this study and variability in geological complexity across different regions. Further testing in diverse depositional environments is required.

Future steps should include integrating these methods into existing workflows within real-world geological operations, facilitating practical adoption by geologists. Specifically, an automated inter-well correlation workflow is planned, addressing the need for scalability and operational efficiency in daily geological analysis. This integration will streamline routine tasks, reduce interpretation times, and improve consistency, significantly benefiting geologists in their day-to-day stratigraphic analysis.

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